

Open-YOLO 3D: Towards Fast and Accurate Open-Vocabulary 3D Instance Segmentation

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Background: 3D scene understanding is transforming the world

- 3D perception powers robotics, AR, and digital twins.
- Instance segmentation enables object-level reasoning and action.
- Open-vocabulary enables naming unseen objects from text prompts.
- Fast, accurate 3D perception is critical for real-time systems.

The problem: Open-vocabulary 3D instance segmentation at speed

- Task: segment 3D instances and label via text prompts.
- Recent solutions rely on SAM and CLIP feature lifting.
- Issue: multi-view 2D→3D aggregation is slow and costly.
- Consequence: unacceptable latency for real applications.

Key challenges in prior art

- Heavy 2D segmentation inflates compute per frame.
- CLIP feature lifting requires dense multi-view fusion.
- Per-instance visibility handled iteratively, not batched.
- Semantic-only scoring often up-ranks bad masks.

- Detector-only prompting: replace SAM+CLIP with OV detectors.
- Low-Granularity label maps: bounding-box fills, not masks.
- VAcc: vectorized, GPU-friendly visibility computation.
- MVPDist: multi-view voting for robust text labeling.
- Joint mask confidence: geometry $\times$ semantics without CLIP.

- 3D proposals, 2D detections, batched visibility, multi-view votes.
- Assign labels via MVPDist and score via $\text{sIoU} \cdot \text{sclass}$.
- No SAM masks, no CLIP feature lifting.

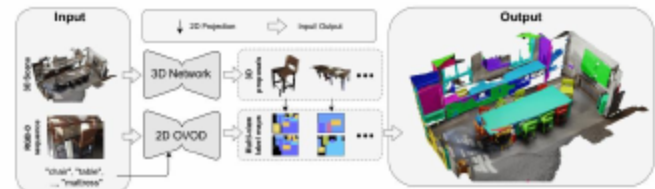


Figure: VAcc assigns top-k visible views; MVPDist aggregates labels at 2D projections to choose the class.

- Takeaway: VAcc + MVPDist turn multi-view loops into batched, GPU-efficient steps.

- Generate K class-agnostic 3D masks per scene.
- Apply NMS to filter overlapping proposals.
- Projections already encode instance structure.
- Avoids costly 2D-to-3D instance fusion.

- Run YOLO-World on each RGB frame.
- Rasterize boxes to frame-sized LG label maps.
- Larger boxes overwrite to mimic occlusions.
- Cheap to store, fast to sample per projection.

- Project all points to all views in one GPU pass.
- Compute in-frame and depth-consistent visibility.
- Aggregate per-mask, per-view visibility scores.
- Select top-k views per mask efficiently.

- Sample LG labels at mask projections in top-k views.
- Aggregate class occurrences into a distribution.
- Assign class with maximum probability.
- Redundancy across views boosts zero-shot robustness.

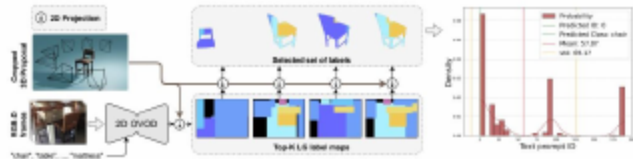


Figure: MVPDist samples labels from top-k visible views and selects the majority class.

- Takeaway: MVPDist picks the most frequent, view-consistent class across visible projections.

- Score: $sm = sloU \times sclass$.
- $sclass$: MVPDist's max class probability.
- $sloU$: projected 3D box vs best 2D box IoU across views.
- Downranks leaking or mislocalized masks.



Figure: $sloU$ = IoU of projected 3D box vs best 2D detector box across views; covers three typical cases.

- Takeaway: $sloU$ grounds semantics in geometry to avoid overconfident but spatially wrong masks.

- LG maps: boxes suffice; 3D masks carry geometry.
- HG via SAM increases latency drastically.
- LG matches HG predictions with less compute.
- Top-k aggregation stabilizes labels further.

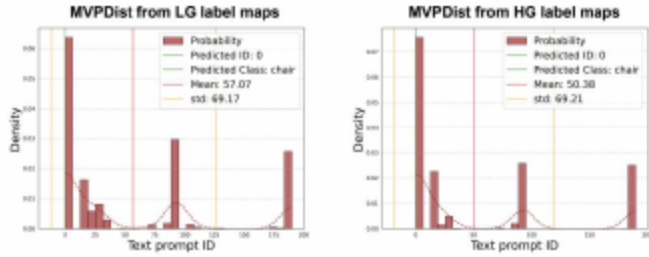


Figure: LG achieves similar predictions to HG with 5× lower runtime.

- Takeaway: LG label maps preserve accuracy while slashing runtime.

- ScanNet200: 312 val scenes, long-tail splits.
- Replica: 48 categories, cross-dataset generalization.
- Metrics: mAP@[0.5:0.95], mAP50, mAP25.
- Frames sampled as in OpenMask3D; single A100 GPU.

- Compare to Open3DIS, OpenMask3D, OpenScene, OVIR-3D.
- Proposal source: Mask3D for fairness and speed.
- Evaluate both accuracy and seconds per scene.
- Two scenarios: oracle masks, proposal masks.

- Open-YOLO 3D: mAP 24.7, mAP50 31.7, mAP25 36.2 at 21.8 s/scene.
- Beats prior CLIP+SAM pipelines in both AP and latency (up to 16× faster).
- Robust across head/common/tail: 27.8/24.3/21.6.

- Open-YOLO 3D: mAP 23.7, mAP50 28.6, mAP25 34.8 at 16.6 s/scene.
- About 11× faster than CLIP+SAM; strong cross-dataset generalization.

Runtime breakdown: where we gain speed

- Visibility and labeling dominate prior runtimes.
- VAcc collapses iterative loops into one pass.
- Detector-only LG maps remove SAM overhead.
- Overall: seconds per scene, not minutes.

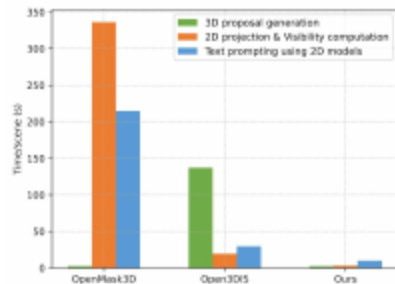


Figure: VAcc removes per-mask loops; detector-only LG maps avoid SAM overhead.

- Takeaway: Batched projection and depth filtering remove per-mask loops.

Replica qualitative results

- Tighter, cleaner instance masks with fewer spurious regions.
- Better adherence to object boundaries; higher precision.
- Consistent under varied lighting and layouts; strong generalization.

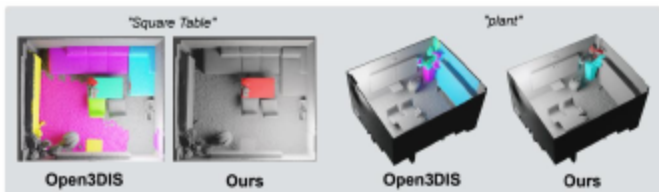


Figure: Joint geometry + semantics improves boundary adherence and precision.

- Takeaway: Joint geometry + semantics improves visual quality across diverse scenes.

Qualitative: open-vocabulary on ScanNet200

- Sharper masks than Open3DIS across prompts.
- Handles occlusions and small objects with improved robustness.
- Zero-shot labels align with text prompts.

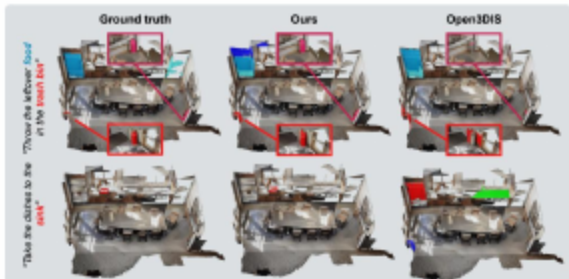


Figure: Multi-view voting stabilizes labels under occlusion and sparse views.

- Takeaway: Multi-view voting adds stability under occlusion and sparse views.

Scoring quality: geometry $\times$ semantics matters

- s_{class} = MVPDist's max class probability; s_{IoU} = 2D box IoU across views from the 3D mask projection.
- Effect: penalizes leakage/mislocalization; promotes well-localized masks.
- Semantic-only methods overvalue bad masks; our ranking elevates well-localized instances.



Figure: s_{IoU} - s_{class} aligns mask scores with human-perceived quality; semantic-only scores overvalue leaky masks.

- Takeaway: s_{IoU} - s_{class} aligns scores with human-perceived mask quality.

- Small-object recall limited by 3D proposals.
- Detector errors can bias MVPDist votes.
- Heavy occlusions reduce informative views.
- Vocabulary coverage depends on detector training.

- We thank collaborators, institutions, and funding agencies.
- Community projects: Mask3D, YOLO-World, and benchmarking datasets.

- How can we boost small-object recall without re-introducing heavy 2D segmentation?
- Can we train end-to-end across detection–projection–labeling for further gains?
- How do we expand vocabulary coverage under domain shift and long tails?
- What's the path to true online/streaming OV 3D segmentation?
- Can lightweight 2D instance cues be fused for better occlusion handling?